Lesson 6: Transformers

ChatGPT edition — use a tiny switched pulse, never mains

Turns ratio under changing flux

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

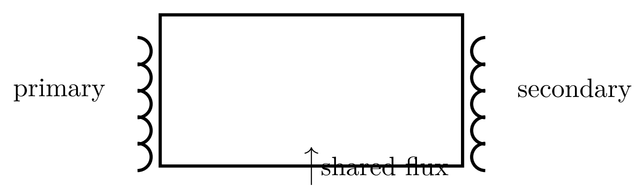
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Two insulated coils on one iron core
- 2 AA cells
- Pushbutton
- Multimeter

Predict first

Will the second coil show a steady voltage while DC in the first coil is steady?



Investigate

1. With power disconnected, put both coils on the core. Connect the secondary only to a voltmeter.
2. Briefly press and release the primary pushbutton. Record both voltage pulses and their signs.
3. Compare secondaries with different turn counts, keeping primary, core, and switching method unchanged.
4. Repeat without the iron core.

Explain from the evidence

A transformer requires changing flux. With DC, change occurs mainly at make and break, hence pulses rather than steady secondary voltage. In the ideal sinusoidal model $V_s/V_p = N_s/N_p$; this crude pulse experiment tests direction and relative magnitude, not precise regulation. Real winding resistance, leakage flux, and core loss matter.

Change one thing

Graph secondary peak against turn count. Do not connect any winding to an outlet or wall adapter.

Evidence record	
Prediction	
One change	
Observation	
Claim + because	

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.